Multi-level Analysis Of The Stability-Flexibility Dilemma

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Abstract

A dominant view in research on cognitive control holds that adjustments of control are constrained by a general stability-flexibility tradeoff. Specifically, increasing the stability of a task-relevant representation makes flexible changes of that representation more difficult (Goschke, 2000b). Although widely replicated, this tradeoff account is typically supported using a single level of analysis of the experimental effects of switch cost as a degree of conflict blockwise (Egner, 2007; Goschke, 2000b, 2000a; Rogers & Monsell, 1995). A smaller body of research supports a different account of stability-flexibility, where on the level of within-subject dynamics stability and flexibility are found to co-occur (Mayr, Kuhns, & Hubbard, 2014). This raises the question that the stability–flexibility tradeoff seen through experimental level effects is not a fundamental property of cognitive control but may instead reflect the analytic level at which it is measured. Moreover, individual differences work hints at substantial variability in how people balance stability and flexibility, suggesting that the relationship may not be uniform across all individuals (Mayr & Grätz, 2024).

The present study investigates where differences in the stability-flexibility dilemma arise through a multi-level analytic approach. Additionally, we take a more direct and sensitive approach to understanding cognitive control mechanisms through looking at the degree of control required pre- and post- task-switching. Across two experiments, we examined stability–flexibility dynamics at three analytic levels: experimental effects, within-subject dynamics, and individual differences. We find that it indeed matters to what level stability-flexibility mechanisms are analyzed. Evidence for the co-occurrence of tradeoff and antitradeoff relationships are found on different levels of analysis, underscoring the importance of a multi-level approach when interpreting stability-flexibility interactions. By showing that stability–flexibility dynamics vary across analytic scales, our results challenge the presumed ubiquity of the dominant tradeoff view and highlight the necessity of multi-level approaches for understanding the mechanisms that govern cognitive control.

Keywords: Cognitive Control, Multi-Level Analysis, Decision-making, Cognition, Task-Switching, Stability-Flexibility, Experimental Effects, Within-Subject Dynamics, Individual Differences

Multi-level Analysis Of The Stability-Flexibility Dilemma

Introduction

A dominant view in research on cognitive control holds that adjustments of control are constrained by a general stability-flexibility tradeoff. Specifically, increasing the stability of a task-relevant representation makes flexible changes of that representation more difficult (Goschke, 2000b). You may be planning to give a speech for example, relying on your notes helps you stay on message reflecting high stability, but it can also make it harder to smoothly switch into an improvised, conversational mode when a question arises thereby reducing flexibility. The very thing that keeps you anchored can also slow you down when you need to pivot. (say this a different way) This potential stability-flexibility tradeoff is also relevant in the clinical world, with impaired regulation of cognitive control being implicated in various neurodevelopmental disorders such as attention deficit hyperactivity disorder, cerebral palsy, and autism spectrum disorder (Cepeda, Cepeda, & Kramer, 2000; D'Cruz et al., 2013; Tajik-Parvinchi, Farmus, Tablon Modica, Cribbie, & Weiss, 2021). In these cases, disruptions in cognitive control can manifest as either overly rigid behavior or excessive distractibility, underscoring the importance of understanding the mechanisms that support both maintaining and updating task-relevant representations.

(explain theoretical model and the fundamental problem with assuming this applies across all levels and peoples) The prediction of a tradeoff relationship arises from connectionist models of cognitive control. The behavior of these models can be described within an "attractor landscape" *(Fig. 1)* (Brown, Reynolds, & Braver, 2007; Geddert & Egner, 2022). In this framework, different task representations form basins in a state space, and the depth of a basin reflects how strongly a task is currently activated. A deeper basin corresponds to a more stable task state, but it also requires more "energy" to move out of, making task switching more difficult. One data pattern that is consistent with this tradeoff prediction arises when examining task-switch costs as a function of previous-trial conflict.

High conflict on the previous trial is assumed to strengthen the currently active task representation, effectively deepening its basin. This increased stability should then make switching on the next trial harder, producing larger switch costs. Although much of the dominant narrative in cognitive control has been built around this tradeoff pattern (Goschke, 2000b), this pattern is surprisingly inconsistent when examining cognitive control at different levels of analysis outside of experimental effects (Brown et al., 2007; Geddert & Egner, 2022; Mayr et al., 2014).

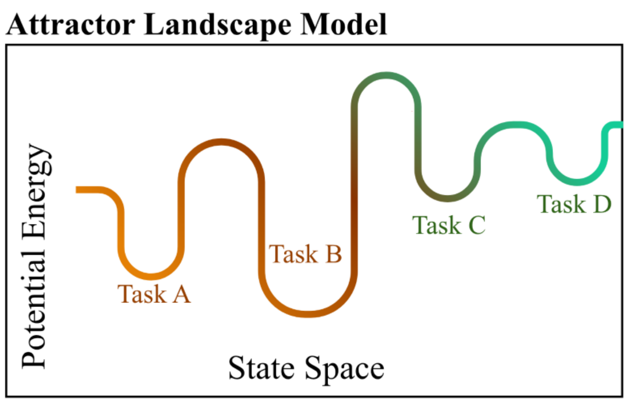

Figure 1. The landscape attractor model. This prediction model represents the dominant view of how stability-flexibility interact, which is in a tradeoff manner. Tasks are represented as basins of varying depth along a state space. The more stable a task representation is, the deeper the corresponding basin in this attractor model. As a result for high stability, it takes more potential energy to move out of the task basin to a different task.

Despite this importance, most empirical work has examined stability–flexibility dynamics using blockwise manipulations, contrasting “stable” versus “flexible” blocks, and has evaluated the tradeoff primarily at the experimental-effects level, averaging across trials and participants (Egner, 2007; Goschke, 2000a; Rogers & Monsell, 1995). (add a bit more)

(blockwise limitation) This blockwise approach has been instrumental in establishing the classic tradeoff pattern, but it also limits what can be observed. Blockwise designs cannot capture moment-to-moment adjustments that occur as people move from one trial to the next, nor can they reveal whether stability and flexibility might co-occur within individuals under certain conditions. As a result, the field may be missing critical dynamics that unfold at finer temporal scales.

(one level manipulation) Additionally, multi-level analytic approach's to understanding stability-flexibility are few and far between, as a result we assume that the dominant tradeoff relationship applies across levels when that is not necessarily the case. From the small body of work that does consider cognitive control on levels other than on the experimental level, we see differences in the tradeoff account, calling into question the generalizability of the dominant relationship (Mayr et al., 2014). Although we cannot conclude that the tradeoff relationship does not also apply also across different levels of analysis, a more rigorous test of cognitive control is needed.

To address these limitations, we take a more sensitive approach to capture switch-cost as a function of trial-by-trial shifts in cognitive control. Manipulating previous-trial to post-switch demands independently provides a more detailed and responsive measure of how stability and flexibility interact as opposed to only understanding stability-flexibility shifts in a blockwise manner. Across two experiments, we use a task-switching 'shape' or 'color' task with the inclusion of univalent (neutral) and bivalent (ambiguous) stimuli in order to understand stability-flexibility as a function of congruence also known as conflict. We examine these dynamics across three complementary levels of analysis: (1) experimental level, (2) within-subject dynamics, and (3) individual differences. On the experimental levels we test whether experimental manipulations of previous and current trial control demands interact with the switch contrast in a manner that is consistent with a stability-flexibility tradeoff model. Then on the within-subjects level we test whether trial-by-trial RT auto-correlations in combination with considering the experimental effects express a stability-flexibility tradeoff. Finally, on the individual differences level we test whether how individuals differ in measures of stability and in measures of flexibility are consistent with the tradeoff model. This multi-level framework allows us to test whether the stability–flexibility tradeoff emerges consistently across analytic level or whether different levels reveal different patterns.

Experimental Level Hypotheses. Here we test whether experimental manipulations of previous and current trial control demands interact with the switch contrast in a manner that is consistent with a flexibility/stability tradeoff model.

No-switch Trials.

H1: Univalent trials (trials where one task is neutral) by definition have no conflict. Therefore, Univalent (Unambiguous) trials result in relatively shorter response times, and fewer errors compared to ambiguous.

H2: Bivalence (ambiguity) enforces stability because it introduces conflict resulting in longer reaction times and errors.

Switch Trials.

H1: Previous trial-group bivalence (ambiguity) enforces stability and should therefore cause higher switch costs (longer response times, more errors). If the previous group of trials was univalent (unambiguous), previous performance required less stability and should therefore reduce switch costs (lower response times, fewer errors).

H2: Current-trial bivalence requires establishing a stronger task set and therefore further increases RT and errors. Previous and current ambiguity should produce particularly large switch costs (e.g., an interaction between previous and current ambiguity).

Within-Subject Level Hypotheses. ()

Individual Differences Level Hypotheses. H1: We test the hypothesis derived from the tradeoff model that participants who are stable under a no-switch condition, which should be more challenging if the previous trials and/or the current trial is bivalent, should be less efficient on switch trials.

Because efficiency in almost all cognitive tasks is positively related (i.e., the "positive manifold") we will conduct a series of model tests, where on each level we will control for relationships occurring on "preceding" conditions to present less of a challenge to the stability-flexibility tradeoff.

H2: We predict that with an increase in stability demands (e.g., non-switch trials where the previous and/or current trial group were bivalent), the coefficients should turn negative in case of a tradeoff.

Method

We report how we determined our sample size, all data exclusions (if any), all manipulations, and all measures in the study.

Participants

Our sample size for experiment 1 was 78, collected using convenience sampling methods. Our sample size for experiment 2 was 100, collected using convenience sampling methods. Both experiments pulled participants from the human subjects pool from the University of Oregon.

Procedure

Participants signed an informed consent form given by our lab, then were instructed through the experiment ahead. Upon learning the instructions, participants completed one practice block, then proceeded to the start of the experiment. Upon completion of the experiment, subjects were debriefed and thanked for their time.

Experiment 1. Tasks and Stimuli. There were 3 possible colors (red, green, blue) and 3 possible shapes (circle, square, and triangle). Additionally, there was a univalent color (gray) and a univalent shape (cross), which made up the fourth stimulus. The univalent stimuli always occurred within the groups of 3 trials. In no condition was a grey cross shown. In each group, the task remained to be univalent or bivalent, with a task switch after each group. Therefore, for each group of 3 trials, the currently irrelevant task could be either univalent or bivalent, and remained the same per group.

Design. The experiment was a task switching paradigm, wherein participants switched between a color and a shape task. In each trial, participants performed one of two tasks- color or shape, where they identify the color or shape of the stimulus, respectively. The task switches every three trials. In other words, the color or shape tasks were active for three consecutive trials. Three trials make up a group. Thus, the first trial of a group marks a task switch into the other task, the following two trials are task repetitions. There was a 100 ms Inter-trial-interval. The participants performed one initial practice block with 126 trials and then moved onto performing 5 experimental blocks with 120 trials each.

Experiment 2. Tasks and Stimuli. The task involved the subject viewing a colored shape. There were 3 possible colors (red, green, blue) and 3 possible shapes (circle, square, and triangle). Additionally, there was a univalent color (gray) and a univalent shape (cross), which made up the fourth stimulus. In no condition both univalent dimensions were shown. Per trial, subjects completed one of two tasks, one task, “color” determined what color the stimuli is and “shape”, determined the shape of the presented stimuli.

Design. Similar to experiment 1, subjects performed one of two tasks indicated by task instructions “color” or “shape”, where they identified either the color or shape attribute of a presented stimulus. The task switched every 4 trials, where the first trial of every group always marked a task switch into another group of 4 trials. Similar to the alternate cues experimental design, the univalent aspect was manipulated randomly in a between trial groups design, in this case with all four trials in one group being bivalent or all four trials in one group being univalent, so that the participants knew that the stimuli alternated task dimension in groups of four. Subjects completed a total of 8 blocks excluding a practice block, each block containing 112 trials.

Measures

first introduce the measure then describe below outcome measure first (iv) predictor measure second

Analysis Plan

We used R (Version 4.5.2; R Core Team, 2025) and the R-packages *dplyr* (Version 1.1.4; **R-dplyr?**), *forcats* (Version 1.0.1; **R-forcats?**), *ggplot2* (Version 4.0.1; **R-ggplot2?**), *hausekeep* (Version 0.0.0.9003; **R-hausekeep?**), *hlmer* (Version 0.1.0; **R-hlmer?**), *janitor* (Version 2.2.1; **R-janitor?**), *lme4* (Version 1.1.38; **R-lme4?**), *lmerTest* (Version 3.2.0; **R-lmerTest?**), *lubridate* (Version 1.9.4; **R-lubridate?**), *Matrix* (Version 1.7.4; **R-Matrix?**), *papaja* (Version 0.1.4; Aust & Barth, 2025), *psych* (Version 2.5.6; **R-psych?**), *purrr* (Version 1.2.0; **R-purrr?**), *readr* (Version 2.1.6; **R-readr?**), *rio* (Version 1.2.4; **R-rio?**), *sjPlot* (Version 2.9.0; **R-sjPlot?**), *stringr* (Version 1.6.0; **R-stringr?**), *tibble* (Version 3.3.0; **R-tibble?**), *tidyr* (Version 1.3.2; **R-tidyr?**), *tidyverse* (Version 2.0.0; **R-tidyverse?**), and *tinylabels* (Version 0.2.5; Barth, 2025) for all our analyses.

Our analysis plan was consistent across both experiments. However, some models we used, particularly when conducting drift-diffusion modeling and in our within-subjects modeling we ran into issues of convergence and/or singularity across both experiments. As a result, we conducted a pre-set series of model comparisons detailed below to achieve the most complex model for each experiment that did not have any convergence or singularity issues.

(write more about our model comparisons decision-tree procedure and how our final models for these ended up looking. can set up here what the procedure was then in each section we describe the ez-ddm models and what our final model looked like, how it differed across exp1 and 2 then again in the within sub section.)

Experimental Level Analysis. Analyses of the dependent variables were conducted on the trial level using multilevel modeling. Independent models for response times and error rates were fit according to the valence (univalent, bivalent) of the previous trial group, the valence of the current trial group, and the switch condition.

(add in figure of the models we outlined in pre-reg?)

These models were run on the data from both experiments. The outcomes of these analyses guided how subsequent analytic layers were approached. We accounted for the possibility of finding evidence for speed–accuracy trade-offs through observation of opposing effects on reaction times and errors but they did not emerge. Consequently, drift-diffusion modeling (EZ-DDM) was applied not as an additional validation of the observed effects and we proceeded with analysis of within-subject dynamics. For the EZ-DDM analysis, the mean and variance of reaction times and mean error rates were calculated for each subject and for the same contrasts specified above. The resulting parameters were analyzed using linear mixed-effects models with random intercepts for each subject to confirm the robustness of the experimental-level findings.

(add in ddm model builds with also explanation across both experiemtns what models were different and why)

Within-Subjects Level Analysis. ()

Individual Differences Level Analysis. Individuals' reaction time and error data were aggregated for all crossings of previous-trial group valence (n-1 bivalent, n-1 univalent), current-trial valence (n bivalent, n univalent), and switch condition (switch trial, no-switch trial). Reaction times were averaged across the no-switch trials within each trial group. Each trial group consisted of three trials for the alternate-cues experiment and four trials for the alternate-runs experiment.

Individual differences analyses of the dependent variable were conducted using linear regression modeling. For each participant, drift rate was computed as a function of switch condition and previous and current group-valence conditions (eight estimates per subject). A stringent test was then performed to determine the presence of a trade-off pattern by adding predictors sequentially. The first model type predicted drift rate in the least-demanding switch condition (univalent-univalent) from the least-demanding non-switch condition (univalent-univalent). Predictors were subsequently added to this model in three steps (see model outline below).

(model outline figure here)

EZ-DDM parameters of non-decision time (Ter) and boundary separation (a) were also computed. If the EZ-DDM non-decision time parameter (Ter) was negative in fewer than 20% of cases across all subjects and conditions, the linear regression models were re-run with data cells in those conditions set as missing. Results obtained from the drift-rate models were cross-referenced with two additional models: one using log-transformed reaction time as the dependent variable and another using error rate as the dependent variable. These models were implemented identically to the steps outlined above.

Results

no interpretations here KISS

figures for experimental effects and for third level effects, prob no vis for within subject level

Discussion

On the Experimental Level.

Inconsistent results across both experiments on the experimental level. Where Altcues (experiment 1): consistent tradeoff relationship occurring in error, logrt and drift Rate. Then on Altruns (experiment 2): inconsistent across variables, logrt and drift rate reveals antitradeoff, error reveals tradeoff relationship.

Alt Cues: lmer & glmer models: higher RTs and more errors when switching into a bivalent task (favoring a tradeoff effect). EzDDM: v is slower when switching into bivalent task (consistent with lmer and glmer)

Alt runs: lmer & glmer models: Alt runs showed support for anti-tradeoff for log RTs and tradeoff for errors. EzDDM: Alt runs ezDDM showed support for anti-tradeoff

On the Within-Subjects Level.

Across both experiments, we see a tradeoff occurring in within-sub dynamics. This is different from the mayrlongterm study mentioned in the review paper that found an anti-tradeoff at the level of within-sub.

Stability-Flexibility Anti-Tradeoff Arises When Examining Within-Subject Dynamics.

note: this needs to be built on when interpreting the data more, read the mayr paper that finds co-occurance when analyzing within-sub dynamics in the meantime. Here we test whether trial-by-trial RT auto-correlations in combination with considering the experimental effects express a flexibility/stability tradeoff.

A recent reevaluation of the generalizability of the stability-flexibility tradeoff has posited that this negative tradeoff relationship may occur only in highly specified contexts (Mayr & Grätz, 2024).These authors pointed toward a possible anti-tradeoff pattern, implying that cognitive stability and flexibility are both driven by a common factor and therefore can co-occur (Mayr et al., 2014). This paper adopts a more nuanced approach by examining when stability–flexibility dynamics emerge, weaken, or disappear. Across two experiments we conduct analysis on the experimental effects, within-subject dynamics, and individual differences level to identify the conditions under which tradeoffs arise and when they do not.

- note:***Cognitive Map Prediction Model.*** Contrary to predictions from the attractor model, there are recent hints suggesting the possibility of an anti-tradeoff pattern, where flexibility and stability appear to co-occur (Mayr et al., 2014; Morales, Moss, & Mayr, 2022). In response to these results, Mayr and Graetz have suggested replacing the attractor-landscape perspective with a cognitive map conceptualization (Fig. 1B). Here, different tasks are locations within a task space that can be represented with varying degrees of resolution. The higher the resolution, the more stable tasks are encoded, but also the easier they are to find when switching, resulting in an anti-tradeoff pattern (Mayr & Grätz, 2024). Despite some evidence for such anti-tradeoff patterns (Morales et al., 2022), this prediction has not been rigorously tested.

On the Individual Differences Level.

Across both experiments, we see an anti-tradeoff relationship occurring on the individual differences level. Co-occurance of stability-flexibility is consistently found across error rate, drift rate and logged RT. Where in general individuals who are better at switch trials will also be better at no-switch trials. Additionally, the switch to no-switch bivalent-bivalent pairing which is the strongest test of switching from stable trial is strongly positive in error rate and drift rate. This means that people who are good at bi-bi switch trials will also be better at bivalent-bivalent no-switch trials.

Here we test whether individual differences in measures of "stability/efficiency" and in measures of flexibility are consistent with the tradeoff model. - tobias egner: individucal dif but looks at data in a block-wise manner. found no relationship - study done in lab reference

note: we find an antitradeoff occuring, need to interpret the results for this section more too. talk about age as a factor or not necessary for this paper? maybe that can come in the discussion portion about other individual dif factors that we can explore with this level of analysis.

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